

David Kreuter

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Senior Infrastructure Engineer | IaC, Observability & Self-Service Platform Builder

Infrastructure engineer with 10+ years of experience designing, building, and operating production systems across cloud and on-premises environments. Architected a global provisioning framework managing 20,000+ virtual machines using Infrastructure as Code, GitOps pipelines, and custom Python middleware. Deep expertise in Terraform, CI/CD automation, observability, and mentoring engineering teams to deliver at scale with governance and operational excellence. Proven ability to drive reliability across geo-redundant datacenters while empowering development teams through self-service platforms and cloud-agnostic abstractions.

WORK EXPERIENCE

M&T Bank

03/2019 - Present

Automation Engineer (formerly Systems Engineer)

- **Hybrid Cloud & GitOps:** Architected GitOps-driven CI/CD pipelines using Terraform to automate the deployment of virtual machines to Azure; implemented a pull-request-based workflow that standardized infrastructure changes through automated plan/apply cycles, establishing repeatable IaC patterns across the organization.
- **Platform Engineering & Middleware:** Designed and developed a custom Python orchestration application as middleware between ServiceNow and Ansible Automation Platform; implemented advanced retry logic, auto-remediation of deployment failures, and a pub/sub queue model for high-volume scaling--delivering a self-service deployment platform for internal teams.
- **Cloud Abstraction & Redesign:** Refactored legacy on-premises automation workflows to be fully cloud-agnostic, enabling seamless deployment to Azure; designed the abstraction layer to be transparent to end users, maintaining a consistent self-service interface while standardizing infrastructure patterns across hybrid environments.
- **Architectural Scale:** Engineered a global server provisioning framework that deployed and managed 20,000+ virtual machines; integrated Rubrik and VMware APIs to automate backup and compute placement, ensuring 100% RTO/RPO compliance and system reliability at build time.
- **Infrastructure Resilience:** Engineered and maintained a highly resilient infrastructure architecture spanning three identical, geo-redundant datacenters to ensure maximum uptime, disaster-recovery readiness, and application SLO adherence.
- **End-to-End Hardening & Provisioning:** Developed a zero-touch build workflow that delivered fully patched and hardened Linux (Ansible) and Windows (GPO) servers; automated agent installation, drive formatting, and configuration management, including optional hardened SQL deployments at scale.
- **Technical Leadership:** Serve as a primary mentor for the Technology Development Program (TDP), setting enterprise-wide standards for naming conventions, Git branching strategies, and unit testing for infrastructure code--shaping infrastructure standards and influencing cross-functional technical decisions.
- **Operational Efficiency:** Developed an intelligent IP provisioning engine via InfoBlox API integrations, eliminating 1,000+ annual engineering hours and resolving a multi-year infrastructure bottleneck through internal tooling that empowered teams to move fast.
- **Network Security Automation:** Automated Check Point firewall rule generation and DMZ provisioning; integrated security-policy updates into the server build cycle to ensure day-zero network isolation, partnering with security and networking teams for full-stack reliability.
- **Database as a Service (DBaaS):** Collaborated with Database Engineering to architect a high-performance on-premises SQL cluster, enabling a unified DBaaS offering that consolidated fragmented legacy databases and streamlined application delivery.
- **Cost Optimization:** Orchestrated the migration of 850 branch file servers within a strict three-month window using custom PowerShell automation, avoiding \$400,000 in extended licensing fees.
- **Risk Mitigation:** Authored Active Directory lifecycle scripts with built-in 'read-only' dry-run modes; implemented least-privilege permissions and robust rollback protocols for stale-object cleanup.

OTG Management

07/2013 - 02/2019

Lead Network Engineer

- **Performance Engineering:** Diagnosed and resolved a mission-critical server hang by identifying a NUMA architecture mismatch; realigned vCPU/memory topology to eliminate cross-node latency and restore system stability--performing root-cause analysis and long-term remediation.
- **Infrastructure Resilience:** Architected a highly redundant DNS solution that significantly reduced network outages and optimized global traffic performance, supporting application reliability and SLOs.
- **Cloud Connectivity:** Established secure, encrypted VPN tunnels to bridge on-premises networks with cloud resources, ensuring seamless hybrid-cloud operations.
- **Security Architecture:** Engineered a secure reverse-proxy solution to protect sensitive internal infrastructure, enabling customer online ordering while mitigating external threat vectors.

Network Engineer

- **Proactive Monitoring:** Developed service-aware alerting and observability frameworks for business-critical systems, driving monitoring strategies that reduced mean time to resolution (MTTR) through proactive incident detection.

- **Network Administration:** Managed complex firewall rules and switch ACLs for a vast distributed network spanning 12 airports and 3 satellite offices, including DNS and packet-level debugging.
- **Cybersecurity Governance:** Directed enterprise-wide vulnerability scanning and remediation efforts for internal and external systems to maintain a hardened security posture.
- **Enterprise Mobility:** Successfully onboarded and managed 15,000 iPads via Meraki MDM, implementing automated security protocols including geofencing, remote-wipe triggers, and custom application deployment.

Systems Administrator

- **Operational Visibility:** Modernized infrastructure monitoring by designing custom Grafana dashboards to visualize real-time performance metrics, driving observability across the infrastructure stack.
- **Disaster Recovery:** Spearheaded the deployment of Datto backup and disaster-recovery appliances to ensure 100% business continuity for critical operations.

SKILLS

Languages: Python, Bash, PowerShell, SQL, YAML, Javascript, PHP

Automation/Orchestration: Terraform, CI/CD Pipelines, GitOps, Ansible Automation Platform, Grafana, Middleware Development, REST API, GraphQL, Gitlab, ServiceNow, Unit Testing

Infrastructure: Infrastructure as Code (IaC), Azure Cloud Services, VMware vSphere, OS Hardening (CIS/STIG), Rubrik, Nutanix AHV, DBaaS, NUMA Tuning, Check Point Firewalls, Active Directory

Networking: DNS, High Availability, BGP, F5 Load Balancing, VPN, VRRP, Intrusion Detection, Vulnerability Scanning, Meraki MDM

EDUCATION

Bachelor Of Science in Computer Information Systems

12/2012

SUNY Fredonia

CERTIFICATIONS

Certified Meraki Network Operator

Meraki

VOLUNTEERING & LEADERSHIP

Habitat For Humanity

Volunteer

07/2025 - Present

Tiftt Nature Preserve

Volunteer

01/2025 - Present

716 Paws

Thermal drone pilot for search and rescue missions.

01/2022 - Present